Venkata Lakshmi Parimala Pasupuleti

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github.com/venkatalakshmiparimala/AgentProbe | New York City Metro Area

PROFESSIONAL SUMMARY

AI Engineer with 9 years of production Python + TypeScript engineering and a live open-source **LLM eval + agent observability toolkit** (AgentProbe) -- trace-level monitoring, 5-metric LLM-as-judge eval engine, ChromaDB RAG quality tracking, OpenTelemetry-compatible tracing semantics, CI/CD quality gates. Built what **Arize Phoenix does**, independently, from first principles. Shipped 20+ production modules at enterprise scale (1.6M+ financial records, 100K+ MAU) and delivered HIPAA-compliant AI on structured healthcare records. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 -- Aug 2025

Research Associate -- Virginia, USA

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries -- multi-source data collection, transformation, and stakeholder reporting.
- Applied Isolation Forest to automate data-quality checks across 3 research pipelines; built Tableau, ggplot2, and Quarto dashboards translating ML outputs into faculty decisions -- 40% comprehension improvement across two cohorts.

Entrata

Jan 2017 -- Aug 2024

Software Engineer

- Applied Python, PyTorch, and OpenAI API to automate AI pipelines across **1.6M financial records** for enterprise property management clients -- production AI on regulated data with output validation, quality gates, and compliance controls.
- Reduced deployment times by 40% building 20+ microservices (Python, PostgreSQL, RabbitMQ) across 120+ Agile sprints for **100K+ monthly active users** at enterprise scale -- 99.5% uptime sustained over 6+ months through direct production ownership.
- Cut microservices execution time 50% via Docker containerization (30%+ of Entrata microservices); improved query performance 30% across 200+ SQL queries; led 300+ peer code reviews mentoring engineers on production AI patterns.
- Achieved 100% PHI compliance migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents; built affordable housing compliance modules (Section 8/HCV, LIHTC, HUD) - income eligibility verification, AMI threshold calculations, and HUD regulatory reporting across 100K+ unit portfolios; built 5+ internal platform tools cutting resolution time 40%.

PROJECTS

AgentProbe: LLM Eval + Agent Observability Toolkit

Open Source

Python · LangChain · LangGraph · FastAPI · ChromaDB · OpenTelemetry · Anthropic · OpenAI · Groq · React · Docker · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

- Built a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing -- deployed as a 15-endpoint FastAPI backend with **OpenTelemetry-compatible trace-level observability**, same architecture as Arize Phoenix, built locally at zero cost.
- Automated GenAI quality gates with a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, semantic retrieval, completeness, compliance) -- failing agent runs blocked before production via CI/CD; React dashboard tracking per-run metrics, regression drift, RAG retrieval quality, and latency across concurrent agent test suites.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker -- github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% building an NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and semantic retrieval -- **HIPAA-compliant RBAC** across structured healthcare records; demonstrates fast AI prototyping in a regulated domain from day one.

PropLens: Lease Intelligence & Compliance AI

In Progress

Python · LangGraph · ChromaDB · FastAPI · LLMs · React

- Building an agent-based RAG pipeline over lease PDFs using LangGraph + ChromaDB vector search -- LLM-as-judge scoring and anomaly detection across 1,000+ tenant records; extends GenAI observability patterns from AgentProbe to a domain-specific compliance use case.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- Jawaharlal Nehru Technological University, Anantapur — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Observability: LLM Observability, GenAI Observability, Agent Tracing, OpenTelemetry, LLM-as-Judge, Eval Frameworks, LLM Evaluation, LangChain, LangGraph, Multi-Agent Orchestration, RAG Pipelines, ChromaDB, Vector Search, Semantic Retrieval, Prompt Engineering, MLOps Pipelines, OpenAI API, Anthropic API, Groq, Hugging Face, FastAPI

ML & Data: PyTorch, TensorFlow, Scikit-learn, Isolation Forest, XGBoost, BERT, spaCy, Embeddings, NumPy, Pandas, Tableau, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, GCP, Docker (production containerization), Kubernetes (orchestration patterns), CI/CD (GitHub Actions), RabbitMQ, PostgreSQL, MySQL